

FIRENZE PERETOLA, Italy

WMO: 161700

Lat: 43.810N

Lon: 11.205E

Elev: 44

StdP: 100.80

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.4	-10.1	1.6	2.8	-8.1	1.9	3.3	9.4	5.7	8.1	6.4	1.0	80	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	35.8	21.8	34.1	21.5	32.8	21.2	24.0	31.0	23.1	30.3	22.4	29.4	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	16.8	26.9	21.0	15.7	26.3	20.1	14.8	25.8	72.0	31.3	68.7	30.6	65.9	29.4	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.9	6.0	-5.8	38.0	1.6	1.7	-6.9	39.2	-7.9	40.3	-8.8	41.2	-10.0	42.5		
(5)			-6.3	25.1	1.5	1.3	-7.4	26.0	-8.3	26.8	-9.2	27.5	-10.3	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	6.7	7.8	10.9	14.4	18.5	23.0	25.6	25.6	21.1	16.8	11.5	7.3
	DBStd	7.37	3.30	3.27	2.96	2.67	2.68	2.91	2.33	2.50	2.83	3.02	3.55	3.67
	HDD10.0	336	109	73	26	2	0	0	0	0	1	26	98	
	HDD18.3	1658	359	295	229	121	30	2	0	0	7	66	207	342
	CDD10.0	2459	8	12	54	133	265	389	483	484	334	211	70	15
	CDD18.3	739	0	0	0	2	37	141	224	226	92	18	0	0
	CDH23.3	7473	0	0	2	46	357	1413	2444	2409	724	77	1	0
	CDH26.7	3261	0	0	0	3	86	586	1180	1178	222	6	0	0
	WSAvg	2.1	1.7	2.1	2.5	2.3	2.3	2.3	2.4	2.3	2.2	1.8	1.8	1.7
	PrecAvg	847	69	70	75	65	65	52	39	63	74	79	117	80
(15) Precipitation	PrecMax	1153	178	190	174	183	126	105	172	129	176	225	236	183
	PrecMin	503	5	1	0	14	0	2	1	0	0	8	10	5
	PrecStd	204	41	48	49	40	39	26	43	39	46	61	61	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	17.8	23.1	26.9	31.2	35.7	37.2	38.1	33.3	27.2	21.6	17.1
		MCWB	11.9	11.0	13.1	16.3	19.3	22.4	21.5	21.7	20.6	18.2	16.5	13.2
	2%	DB	14.5	15.9	20.1	24.8	29.0	33.9	35.7	36.0	31.2	25.2	19.2	15.4
		MCWB	11.1	10.4	12.1	15.4	18.7	21.8	21.6	21.7	20.0	18.2	15.0	12.4
	5%	DB	13.1	14.2	18.2	22.6	27.0	32.1	34.0	34.1	29.1	23.9	18.0	14.1
		MCWB	10.0	9.8	11.7	14.4	18.0	21.1	21.4	21.6	19.4	17.4	14.2	11.4
	10%	DB	12.0	13.1	16.8	20.8	25.1	30.2	32.2	32.5	27.2	22.1	16.8	12.8
		MCWB	9.3	9.5	11.2	13.8	17.0	20.5	21.0	21.2	18.7	17.0	13.7	10.3
	0.4%	WB	13.2	12.7	14.7	17.7	20.9	24.6	24.8	25.4	22.7	20.6	17.5	14.3
		MCDB	15.1	15.3	19.2	24.1	28.0	32.8	32.0	32.4	28.7	24.3	20.3	16.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.8	11.7	13.5	16.2	19.6	23.1	23.7	24.2	21.6	19.6	15.8	13.1
		MCDB	13.7	14.0	17.7	22.4	26.7	30.9	31.0	30.9	27.6	23.1	18.2	15.0
	5%	WB	10.9	10.8	12.7	15.2	18.7	22.1	22.9	23.3	20.8	18.6	15.0	12.0
		MCDB	12.7	13.3	16.5	20.8	25.2	29.6	30.4	29.9	26.7	22.1	17.4	13.7
	10%	WB	9.9	10.1	11.9	14.3	17.7	21.1	22.2	22.5	19.9	17.6	14.0	10.8
		MCDB	11.3	12.4	15.6	19.4	23.7	28.3	29.5	29.2	25.6	21.1	16.2	12.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	9.7	10.8	11.2	12.0	12.6	12.8	11.2	9.6	7.9	7.5
		MCDBR	8.8	10.9	13.3	14.8	14.6	14.8	15.1	15.4	14.0	11.4	8.9	8.2
		MCWBR	5.8	6.5	6.9	7.1	6.4	5.4	4.9	4.9	5.0	5.3	5.3	5.5
		MCDBR	6.8	8.2	10.2	12.4	12.6	13.1	12.7	12.5	11.4	9.1	7.6	6.6
	5% WB	MCWBR	5.0	5.4	5.7	6.3	5.8	5.4	4.8	4.8	4.8	4.5	4.8	4.8
(40) Clear-Sky Solar Irradiance	taub		0.344	0.368	0.406	0.433	0.438	0.444	0.434	0.427	0.414	0.405	0.375	0.342
	taud		2.420	2.328	2.230	2.181	2.208	2.237	2.270	2.305	2.320	2.320	2.345	2.440
	Ebn at Noon		772	813	824	831	838	832	835	828	808	759	720	738
	Edh at Noon		77	99	123	140	140	137	131	122	112	99	82	69
(44) All-Sky Solar Radiation	RadAvg		1.49	2.30	3.51	4.68	5.83	6.70	6.92	5.97	4.35	2.75	1.61	1.29
	RadStd		0.19	0.32	0.51	0.45	0.54	0.38	0.39	0.38	0.25	0.30	0.18	0.16

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	+1.11	-0.84	-0.97	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70	N/A

Nomenclature: See separate page